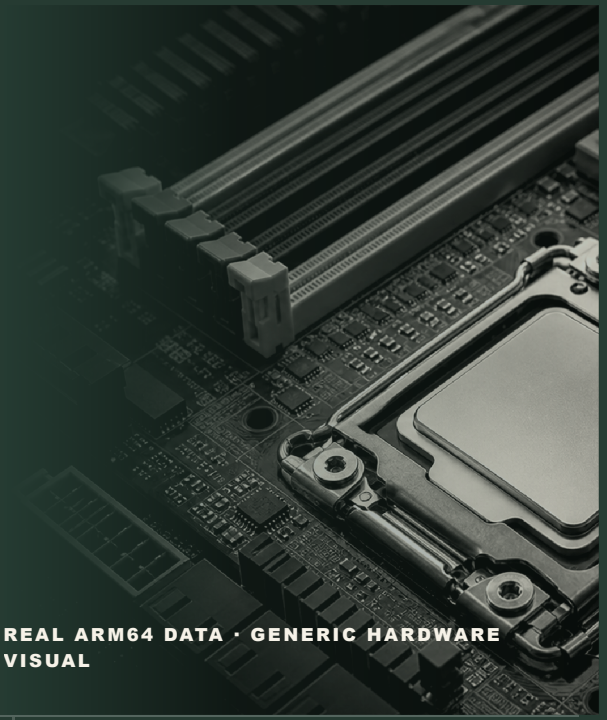
ARM AI OPTIMIZATION CHALLENGE · CLOUD AI

PerfRunbench

Guarded autotuning for repeatable AI workloads

Bounded search, statistical confirmation, and safe recommendation reuse.



REAL ARM64 DATA · GENERIC HARDWARE VISUAL

+86.5%
output throughput

−98.0%
p95 TTFT

608
measured requests · 0 failures

C7g Arm64 server · T4g Arm64 remote client · Qwen2.5-7B Q4_K_M · concurrency 4 · direct private path

01 · THE ACTUAL PROBLEM

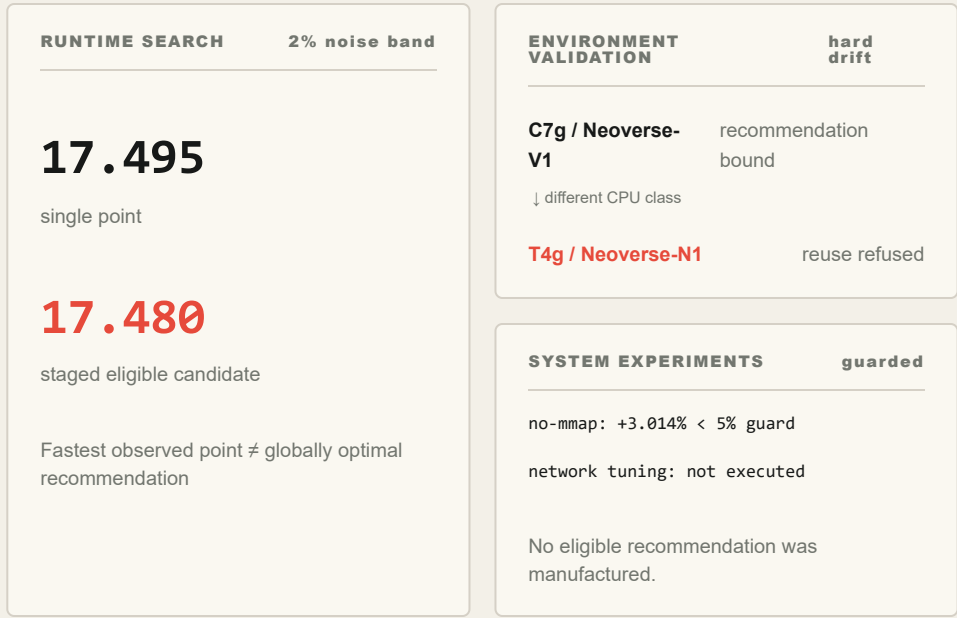
The fastest run is not automatically the safest recommendation.

The best setting depends on the target, workload, runtime, and measurement protocol.

The fastest run may be noise.

The environment may drift.

The apply step may be unsafe.



02 · PRODUCT INTERACTION

One command starts a bounded, process-scoped tuning run.

CONFIG-FREE ENTRY

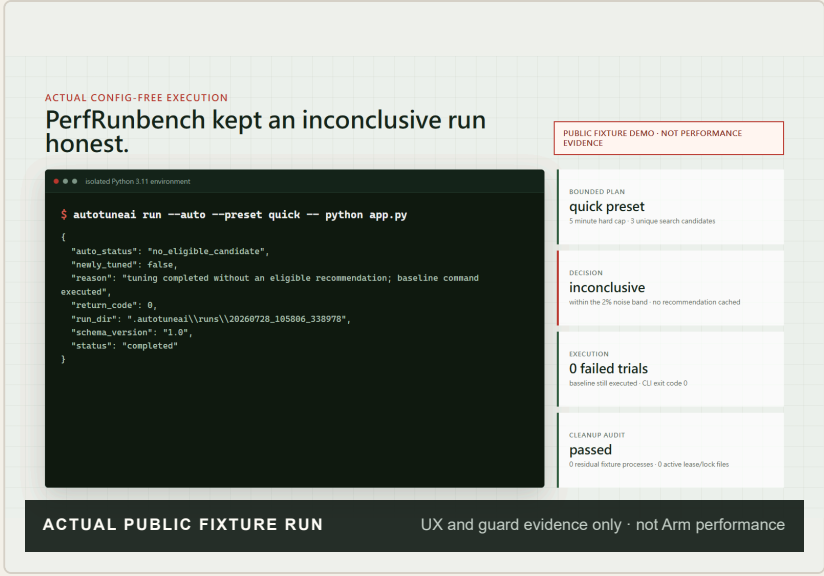
```
$ autotuneai run --auto --preset quick -- python app.py
```

quick 5 min / 3 candidates **standard** 20 min / 8 candidates **deep** 60 min / 16 candidates

01 **No configuration file**

02 **Hard execution budget**

03 **No machine-wide state change**



When the hard budget expires, PerfRunbench still writes a valid partial report and runs cleanup; invalid evidence is not cached.

03 · GUARDED PIPELINE

Every recommendation carries the evidence needed to reuse it safely.

01 **Fingerprint**
Command, config, target, runtime

02 **Search**
Deterministic plan, hard budget

03 **Repeat**
Warm-up exclusion, paired runs

04 **Confirm**
Noise band, failures, sample guard

05 **Passport**
Digests, provenance, statistics

06 **Guarded reuse**
Validate, canary, lease, rollback

Measure once. Reuse only when identity still matches.

- × parser failure
- × inside noise band
- × hard drift
- × cleanup failure

04 · REMOTE SERVING THROUGHPUT

69.4–86.5% more output throughput.

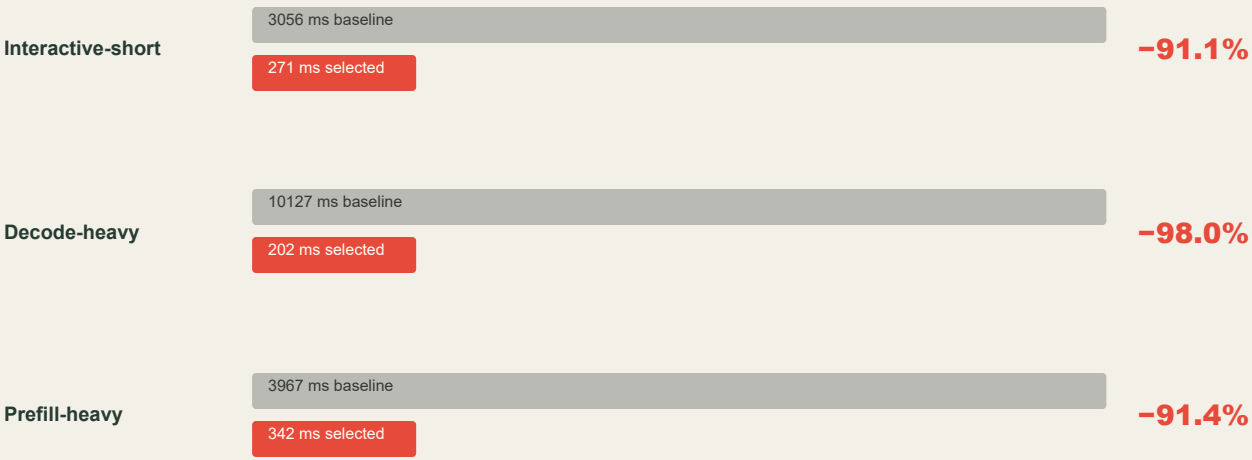


Selected runtime 4 parallel slots · continuous batching on · ctx 2048 · batch 512 · ubatch 128

C7g server · T4g client · direct private path · concurrency 4

05 · LATENCY + RELIABILITY

p95 time-to-first-token fell by 91.1–98.0%.



608

measured requests

0 retries

0 request failures

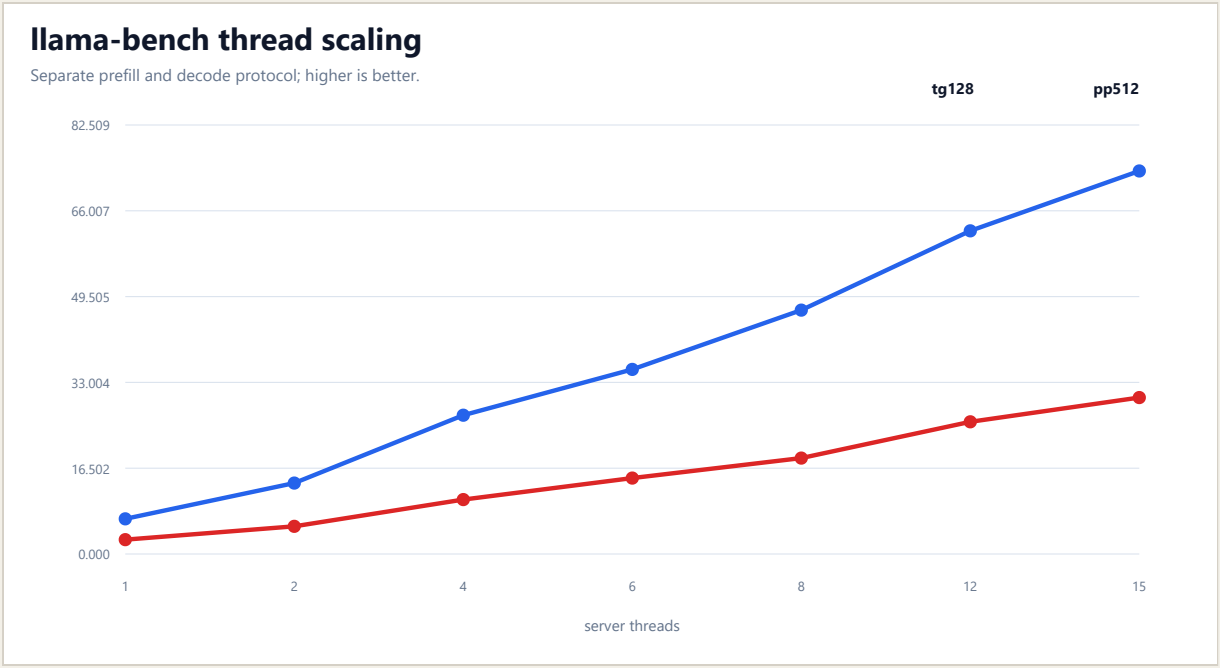
0 quality failures

Artifacts validated · both hosts restored

Client-observed TTFT · fresh server restart between blocks · deterministic randomized block order · same concurrency and workload protocol per comparison.

06 · LARGER ARM CAPACITY

Fifteen Arm cores delivered 10.9× single-stream throughput.



PREFILL · PP512

6.761 → 73.669

10.897×

DECODE · TG128

2.759 → 30.089

10.907×

72.6–72.7%

parallel efficiency at 15 threads

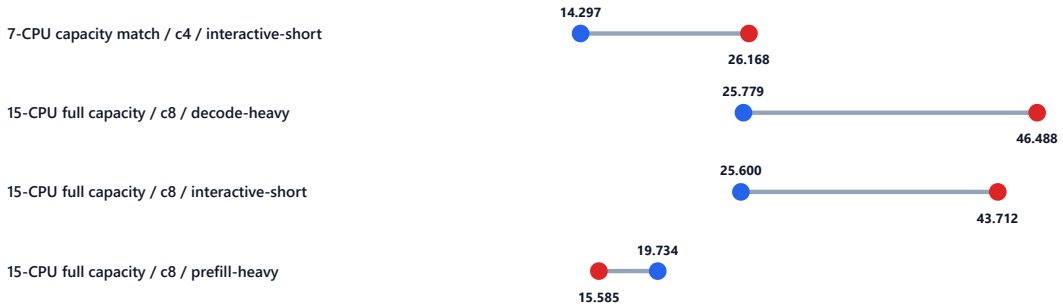
Separate llama-bench protocol · one C7g.4xlarge · five measured repetitions per point · increasing shared-resource pressure is inferred, not proven by memory-controller counters.

07 · WORKLOAD-AWARE RECOMMENDATION

One profile helped two workloads and hurt another.

Serving before and after

Each row compares one paired protocol; runtime and capacity stages stay separate.



+80.3% decode-heavy · eligible

+70.7% interactive-short · eligible

−21.0% prefill-heavy · rejected

8-SLOT CONTEXT GUARD

412 required > 256 available

HTTP 400 failures preserved. No metric was invented.

Full-capacity loopback · concurrency 8 · three randomized paired blocks · fresh server restart per side · only interactive and decode enter the recommendation.

08 · STATISTICAL GUARD

A trustworthy tuner must be able to say no.

Measured ≠ eligible

PerfRunbench recommends only candidates that clear evidence, provenance, correctness, and cleanup gates.

NO ELIGIBLE RECOMMENDATION

Later compiler / backend rerun	inside 5% noise band
no-mmap	+3.014% · below guard
4K / 64K kernel profiles	no qualified improvement
Network tuning	not executed · no bottleneck evidence

Negative evidence is preserved. Unsupported, regressed, or inconclusive candidates remain visible instead of becoming a speedup claim.

09 · ARM RELEVANCE + PORTABILITY

Performance evidence and compatibility validation are separate claims.

MEASURED ON ARM64

AWS C7g

Arm Neoverse-V1 server

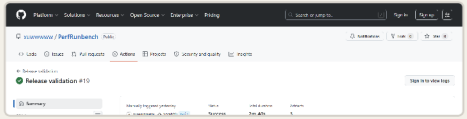
C7g server —— direct private path —— **T4g client**

- Qwen2.5-7B Q4_K_M
- pinned llama-server binary
- 608 remote measured requests

COMPATIBILITY VALIDATED

Ubuntu	Arm64	**PASS**
Ubuntu	x86-64	**PASS**
Windows	x86-64	**PASS**

Installed-wheel CI validates packaging and behavior. It is not a performance benchmark.



Release validation · all three runners succeeded

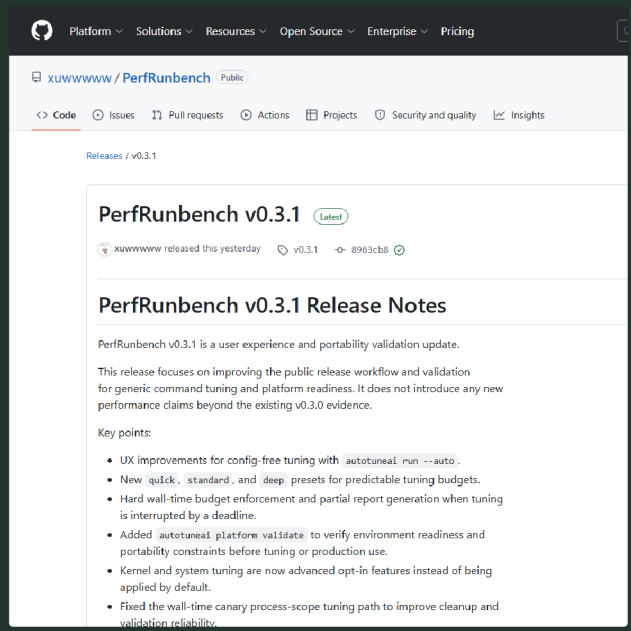
GitHub Actions run 30234620475 · commit 5ccab60

PERFRUNBENCH · v0.3.1

Measure once. Reject false gains. Reuse safely.

Best observed eligible configuration under the tested target, model, workload, runtime, concurrency, topology, and measurement protocol.

REPOSITORY	github.com/xuwwwww/PerfRunbench
RELEASE	github.com/xuwwwww/PerfRunbench/releases/tag/v0.3.1



Published GitHub Release · verified wheel + sdist digests

@xuwwwww · Arm AI Optimization Challenge · Cloud AI

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